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Suspension culture promotes serosal mesothelial development in human intestinal organoids

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SUMMARY

Pluripotent-stem-cell-derived human intestinal organoids (HIOs) model some aspects of intestinal development and disease, but current culture methods do not fully recapitulate the diverse cell types and complex organization of the human intestine and are reliant on 3D extracellular matrix or hydrogel systems, which limit experimental control and translational potential for regenerative medicine. We describe suspension culture as a simple, low-maintenance method for culturing HIOs and for promoting *in vitro* differentiation of an organized serosal mesothelial layer that is similar to primary human intestinal serosal mesothelium based on single-cell RNA sequencing and histological analysis. Functionally, HIO serosal mesothelium has the capacity to differentiate into smooth-muscle-like cells and exhibits fibrinolytic activity. An inhibitor screen identifies Hedgehog and WNT signaling as regulators of human serosal mesothelial differentiation. Collectively, suspension HIOs represent a three-dimensional model to study the human serosal mesothelium.

Graphical abstract

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AUTHOR CONTRIBUTIONS

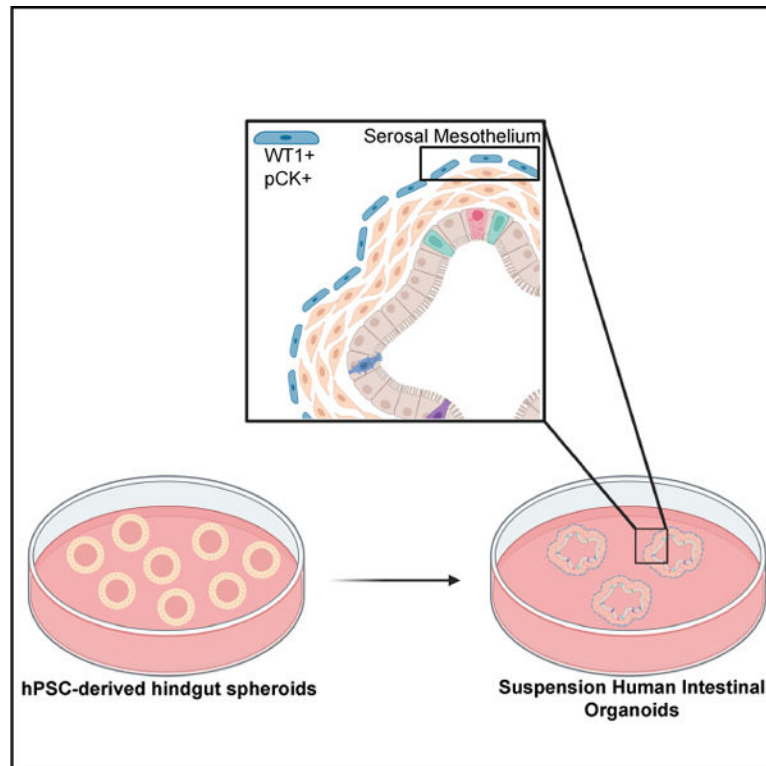
Project conceptualization, M.M.C., S.H., and J.R.S.; experimental design, M.M.C., J.R.S., S.H., and Y.-H.T.; experiments and data collection, M.M.C., S.H., Y.-H.T., A.W., N.G., E.M.H., N.S., and C.B.; data analysis and interpretation, M.M.C., C.C., J.H.W., S.H., Y.-H.T., A.W., and N.G.; writing manuscript, M.M.C. and J.R.S.; editing manuscript, all authors.

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2022.110379>.

DECLARATION OF INTERESTS

The authors declare no competing interests.



In brief

Capeling et al. describe suspension culture for human intestinal organoids as an alternative to Matrigel. Suspension organoids develop a serosal mesothelial layer that resembles the serosa of the human fetal intestine. Using this system, HH and WNT signaling are implicated in differentiation and patterning of the human intestinal serosal mesothelium.

INTRODUCTION

Human intestinal organoids (HIOs) are 3D tissues derived from human pluripotent stem cells (hPSCs) that mimic the structure and function of the human intestine (McCracken et al., 2011; Spence et al., 2011). However, HIOs lack some key cell types found in the native intestine, including vasculature and neurons (Finkbeiner et al., 2015; Holloway et al., 2020; Schlieve et al., 2017; Watson et al., 2014; Workman et al., 2016). HIOs offer advantages over 2D cell-based model systems that do not recapitulate the 3D architecture of human organs or animal models that do not always mimic human physiology (Dedhia et al., 2016). While HIOs are a promising tool to study intestinal development and disease, they have been hindered by reliance on basement membrane extracellular matrix (ECM) products (i.e., Matrigel). These ECMs support growth but introduce biological variability and are not amenable to clinical applications due to high cost and xenogeneic origin (Czerwinski and Spence, 2017). We have recently demonstrated that HIOs, which possess both epithelium and mesenchyme, create their own basement membrane and can thus be cultured in biologically inert alginate hydrogels (Capeling et al., 2019). The observation that HIOs can grow in the absence of a biochemically supportive ECM led us to hypothesize

that HIOs may not require extrinsic support. Here, we demonstrate that a 3D substrate is dispensable for HIO culture and that HIOs can be cultured in suspension. This technique increases the simplicity of HIO culture, streamlines maintenance while enhancing the ability to scale up, and reduces experimental cost and variability. Notably, while HIOs cultured in suspension possessed expected epithelial cell types, they exhibited enhanced mesenchymal organization, including a serosal mesothelial-like layer.

The serosal mesothelium is the outermost layer of the intestine composed of a single layer of squamous mesothelial cells (Bloom and Fawcett, 1975). It provides a protective boundary for the intestine and creates a lubricating, frictionless surface (Mutsaers, 2004) that is involved in homeostasis and disease (Brida et al., 2007; Dobbie, 1992; Federici et al., 2003; Pumberger et al., 2002; Sulaiman et al., 2001; Waldhausen and Sawin, 1997; Winters and Bader, 2013). The serosa also plays a critical role in development via contributions to mesenchyme, especially vascular smooth muscle, in multiple organs (Carmona et al., 2013; Dettman et al., 1998; Que et al., 2008; Wilm et al., 2005). Despite the need to better understand this tissue, a serosal mesothelium has never been described in complex 3D models of the human intestine.

We compared suspension HIOs with the developing human intestine and show that the serosal mesothelial layer within suspension HIOs is similar to that of the human intestine at the cellular, molecular, and functional levels. Using suspension HIOs, we interrogated signaling pathways that control mesothelial differentiation. By carrying out a targeted inhibitor screen, our results implicate Hedgehog (HH) and WNT signaling as a key regulator of serosal mesothelial formation in the human intestine. Overall, these studies introduce added complexity to the HIO model system in order to study development of the intestinal mesenchyme and serosa.

RESULTS

HIOs grow and mature in suspension culture

Based on our observation that HIOs can be cultured in unmodified alginate, a bioinert hydrogel that provides purely mechanical support (Capeling et al., 2019), we hypothesized that the 3D support provided by ECM may be dispensable for HIO development. We tested this hypothesis using suspension culture as an alternative method to grow HIOs. Suspension culture has been utilized in other organoid systems (Hohwieler et al., 2017; Kumar et al., 2019; Takahashi et al., 2018) and is a simple, cost-effective method that is amenable to scale-up. Intestinal hindgut spheroids were generated using a previously described method (Capeling et al., 2020; Finkbeiner et al., 2015; McCracken et al., 2011; Spence et al., 2011). Instead of transferring spheroids to a 3D droplet of alginate or Matrigel, we transferred spheroids to a low attachment plate containing HIO growth media (Figure 1A). By 4 weeks of culture, suspension HIOs resembled alginate and Matrigel HIOs as assessed by bright-field microscopy and hematoxylin and eosin (H&E) staining, with a defined inner epithelium and outer mesenchyme (Figures 1A and 1B).

We used histological techniques to examine the effects of suspension culture. HIOs cultured in suspension developed an inner ECAD⁺ epithelium surrounded by VIM⁺ mesenchyme.

The epithelium of suspension HIOs expressed the intestinal transcription factor CDX2 as well as the duodenum marker PDX1. In addition, suspension HIOs had a properly polarized epithelium, as the tight junction marker ZO-1 was expressed across the apical surface (Figure 1C). Suspension HIOs expressed the proliferation marker KI67 throughout the epithelium and mesenchyme, demonstrating that HIOs are able to proliferate in suspension. Similar to alginate and Matrigel HIOs cultured *in vitro*, the epithelium of suspension HIOs was largely immature, as evidenced by broad expression of the progenitor marker SOX9 (Hill et al., 2017). However, suspension HIOs gave rise to differentiated intestinal epithelial cell types, including DPP4⁺ enterocytes, MUC2⁺ goblet cells, and CHGA⁺ enteroendocrine cells (Figure 1D; Capeling et al., 2019). To directly compare suspension HIOs with alginate and Matrigel HIOs, we performed qRT-PCR and found that expression levels of most markers tested displayed no significant differences between conditions (Figure S1B). While HIOs are relatively immature *in vitro* (Finkbeiner et al., 2015; Yu et al., 2021), we found that suspension HIOs undergo maturation when transplanted into mice, as has been described for alginate and Matrigel HIOs (Capeling et al., 2019; Watson et al., 2014; Figures S1B–S1E). Together, these results demonstrate that suspension culture supports the development of HIOs and gives rise to an epithelium that resembles alginate and Matrigel HIOs *in vitro* and *in vivo*.

HIOs cultured in non-adherent conditions form putative serosal mesothelium

While suspension HIO epithelium was similar to alginate and Matrigel HIOs, we observed differences in mesenchymal organization. In alginate and suspension HIOs, the mesenchyme became radially oriented in a manner that more closely resembled the fetal intestinal mesenchyme. This may be due to the non-adherent nature of alginate and suspension culture, since mesenchymal cells were unable to interact with a surrounding matrix and spread away from the epithelium as in Matrigel. In order to characterize mesenchymal differences across culture conditions, we performed H&E staining. Notably, we observed an outer cell layer in alginate and suspension HIOs that resembled the serosal mesothelium of the developing human intestine (Bloom and Fawcett, 1975; Mutsaers, 2002), while a defined outer layer was not observed in Matrigel HIOs (Figure 1D).

In order to confirm serosal identity in alginate and suspension HIOs, we performed immunostaining for mesothelial markers. Wilms' tumor protein (WT1) and cytokeratin proteins are well-characterized markers of mesothelial cells (Chan et al., 1988; Foley-Comer et al., 2002; Kumar-Singh et al., 1997; Watt et al., 2004; Wilm et al., 2005; Winters and Bader, 2013). In addition, previous studies have indicated that the serosal mesothelium sits on a laminin-rich basement membrane (Mutsaers, 2002; Winters and Bader, 2013). In Matrigel HIOs, a serosal mesothelium was never observed as WT1 was dispersed throughout the mesenchyme, and Matrigel HIOs did not form an outer basement membrane (Figure 1E). In both alginate and suspension HIOs, we observed a putative serosal mesothelium marked by co-expression of WT1 and pCK and lined by a LAM⁺ basement membrane (Figure 1E). Mesothelial staining patterns in alginate and suspension HIOs resembled the serosal mesothelium of the human fetal intestine. In addition, suspension HIOs formed a microvillus-lined surface typical of mesothelial cells based on transmission electron microscopy (TEM) and immunostaining for the microvillus marker VIL1 (Figure S2A;

Mutsaers, 2004). We assessed levels of the secreted fibrinolytic agent, t-PA, which mediates fibrinolytic activity of the serosa, and found that suspension HIOs had higher levels of t-PA secreted into the media than Matrigel HIOs (Figure S2B). This suggests that HIO serosa exhibits some expected functions of human serosal mesothelium, including fibrinolytic activity and microvillus formation (Mutsaers, 2004; Ziprin et al., 2003).

HIO serosa resembles human serosa at the molecular level

To characterize the developing human serosa and more closely compare HIO serosa with human fetal serosa, we analyzed published single-cell RNA sequencing (scRNA-seq) human fetal intestine data and generated scRNA-seq data for suspension HIOs (Holloway et al., 2020, 2021; Figures S2C and S2D). From the human fetal data, we identified a small subset of cells within mesenchymal cluster 8 that expressed mesothelial markers *WT1* (Rudat et al., 2014), *UPK3B* (Kanamori-Katayama et al., 2011), *MSLN* (Rinkevich et al., 2012), and *KRT19* (Lua et al., 2015; Figure S2E). In order to better define the subset of human fetal serosal cells, we computationally extracted and re-clustered 761 cells from cluster 8 to identify a subset of cells (sub-cluster 2) as human serosal mesothelium based on expression of *WT1*, *UPK3B*, *MSLN*, and *KRT19* (Figure 2A). We similarly performed scRNA-seq on suspension HIOs, which displayed expected epithelial (cluster 4) and mesenchymal (clusters 0, 1, 2, and 3) cell lineages (Figure 2B). We identified cells within cluster 2 that expressed mesothelial markers (Figures 2B and S2F). We computationally extracted cluster 2 and performed sub-clustering on 989 cells to identify sub-cluster 0 as a serosa-like population expressing the highest levels of *WT1*, *UPK3B*, *MSLN*, and *KRT19* (Figure 2C).

From the extracted serosa sub-clusters within human intestine and suspension HIOs, we obtained lists of the most differentially expressed genes defined as a log₂ fold >1.5 in expression of a gene in the serosa cluster relative to all other clusters (Table S1). The human fetal and HIO serosa shared 45 genes in common: 18.6% of all genes or 31.0% and 31.6% of genes in each list, respectively—a statistically significant overlap (Figure 2D). When comparing any cluster with all other clusters (fetal intestine, suspension HIO, fetal serosa, and HIO serosa), the average overlap between gene sets is 2.4%. This suggests that comparing *in vivo* with *in vitro* serosa has a 7.8-fold increase in gene expression overlap than would be expected by chance. The list of genes upregulated in both human and HIO serosa revealed well-documented mesothelial genes and markers that are not well defined in the literature, including *CAV1*, *CAV2*, and *EZR* (Figure 2D). Histological analysis confirms that these markers are expressed in human fetal and HIO serosa (Figure S2G). This analysis provides further confirmation that HIO serosa resembles human intestinal serosa and identifies previously understudied genes that mark this population.

HIO serosa retains functional capability to differentiate into smooth-muscle-like cells

Mesothelial cells have a well-described function of giving rise to mesenchymal cells, including vascular smooth muscle (Carmona et al., 2013; Que et al., 2008; Wilm et al., 2005). In addition, previous studies have demonstrated that isolated mesothelial cells undergo differentiation into smooth-muscle-like cells *in vitro* (Kawaguchi et al., 2007; Wada et al., 2003). In order to test whether HIO serosa exhibits the expected functionality of differentiating into smooth-muscle-like cells, we devised a fluorescence-activated cell

sorting (FACS) strategy to enrich serosal cells. We observed that human and HIO serosa express PDPN and ECAD, but not EPCAM (Figures 2E and S3A). This allowed us to separate serosa (PDPN⁺/ECAD⁺/EPCAM⁻) from epithelium (PDPN⁻/ECAD⁺/EPCAM⁺) and non-serosal mesenchyme (PDPN⁻/ECAD⁻/EPCAM⁻; Figure S3B). We performed qRT-PCR on the sorted cell populations and verified that serosal mesothelial genes, including *WT1* and *UPK3B*, were significantly enriched in the sorted PDPN⁺/ECAD⁺/EPCAM⁻ population (Figures 2F and S3C).

Sorted HIO serosa cells were cultured *in vitro* for 24 h as an initial time point (isolated HIO serosa) and compared with sorted cells cultured for 7 days. We observed a significant increase in the percentage of cells that were α SMA⁺ after 7 days compared with the 24-h time point, as 30.8% $\pm$ 6% of cells expressed α SMA after 7 days (Figures 2G and 2H). We additionally performed qRT-PCR to compare freshly sorted HIO serosa with HIO serosa cells cultured *in vitro* for 7 days (Figure 2H) and observed increased expression of smooth muscle markers *ACTA2* and *TAGLN* (Figure 2I). These results demonstrate that HIO serosa behaves in a manner typical of mesothelial cells differentiating into smooth-muscle-like cells.

HIO serosa formation is enhanced in suspension culture

Our data have suggested that suspension and alginate HIOs possess an organized serosa-like layer, whereas Matrigel HIOs sometimes possess few disorganized WT1⁺ cells. In order to further interrogate similarities and differences, we combined scRNA-seq analyses of 28-day HIOs cultured in Matrigel, alginate, and suspension. We observed that cells from all three conditions were represented in every cluster (Figure 3A). The proportion of cells that were epithelial was similar between culture conditions, while mesenchymal populations contributed different proportions (Figures 3A and S4B–S4D). From this combined analysis, we identified a population within cluster 3 that expressed mesothelial markers (Figure 3A). We computationally extracted and re-clustered this population, revealing that subcluster 1 was enriched for *WT1*, *UPK3B*, *MSLN*, and *KRT19* (Figure 3B). Within sub-cluster 1, 82.8% of cells originated from suspension HIOs, with 11.0% and 6.2% from alginate and Matrigel HIOs, respectively (Figure 3C). We additionally compared the distribution of cells within each sub-cluster across conditions and found that 61.5% of suspension HIO cells in cluster 3 fell into the mesothelial sub-cluster 1, compared with 31.6% and 6.7% in alginate and Matrigel, respectively (Figure 3D). This analysis suggests that serosa formation is enhanced in suspension culture.

To further quantitate HIO serosa formation across conditions, we calculated the percentage of HIOs cultured in suspension, alginate, and Matrigel that formed a serosa. We observed the highest frequency of serosa formation in suspension culture and observed that serosa formation decreased as alginate density increased (Figure 3E). Similarly, we observed high expression of the mesothelial markers *WT1* and *UPK3B* by qRT-PCR in suspension culture with decreased expression in alginate HIOs and little to no expression in Matrigel HIOs (Figure 3F). If serosa formation was driven solely by lack of adhesion, we would expect to see no differences based on alginate gels of any concentration, given that alginate gels of increasing polymer concentration exhibit a higher storage modulus (Capeling et al.,

2019). These data suggest that compressive or attractive forces from the ECM may hinder mesothelial development, while the absence of compressive forces in suspension culture promotes organization of a serosal mesothelium.

Hedgehog and WNT signaling are implicated in serosal mesothelial development

Based on similarities between human and HIO serosa, we utilized suspension HIOs to investigate signaling pathways involved in serosa formation, since little is known about mesothelial cell differentiation in the human intestine (Thomason et al., 2012; Winters et al., 2012). Prior to interrogating signaling pathways involved in serosa formation, we determined when serosa first differentiates by analyzing an scRNA-seq time course on HIOs (Holloway et al., 2020) as well as immunofluorescent staining (Figures 4A, 4B, and S3F). We found that HIO serosa formation begins after day 7 of culture and is complete between days 14 and 28. Using this developmental timeline, we applied activators and inhibitors of major signaling pathways known to be involved in mesothelial differentiation or intestinal mesoderm development (Colunga et al., 2019; Dixit et al., 2013; Inagaki et al., 2019; Kolterud et al., 2009; Kruithof et al., 2006; Schlueter et al., 2006; Tian et al., 2015; VanDussen et al., 2012; Verzi and Shivdasani, 2008) using suspension HIOs and assessed their effects on the serosa.

Suspension HIOs were grown for 7 days before applying signaling activators and inhibitors to the culture medium so as not to interfere with early developmental patterning. Starting on day 8, we treated suspension HIOs with inhibitors of the fibroblast growth factor (FGF) (SU5402), BMP (Noggin), Notch (DAPT), WNT (IWR1), and HH (cyclopamine) signaling pathways. To assess serosa differentiation, we devised a qualitative grading scheme to score HIOs based on staining for WT1 and pCK. Scoring ranged from 0 to 3, where 0 indicated no serosa, 1 indicated a partial WT1⁺pCK⁺ outer serosa, 2 indicated a complete or “perfect” serosa with WT1⁺pCK⁺ staining surrounding the HIO, and 3 indicated a full or partial outer serosa with additional ectopic WT1 staining in the mesenchyme (Figure 4C).

In basal media, 45.5% ± 6% of HIOs were scored “2,” a complete serosa (Figure 4D). We compared the percentage of HIOs with score 2 in basal media with inhibitor-treated HIOs to determine whether blocking key signaling pathways led to a change in the frequency of HIO serosa formation. Inhibiting FGF, BMP, or Notch signaling did not significantly alter the percentage of HIOs that formed a complete serosa (Figures 4E and S5A). On the other hand, inhibition of WNT or HH signaling with IWR1 or cyclopamine, respectively, led to a significant decrease in the percentage of HIOs with score 2 (Figure 4E).

To follow up on these findings, we pursued HH and WNT modulation experiments. Inhibition of HH signaling with cyclopamine in suspension HIOs led to a majority of HIOs with score 3, indicating that HH inhibition led to excess WT1 expression throughout the mesenchyme, while activation of HH signaling with SAG resulted in a majority of HIOs scoring 2 (Figures 4F and S5B). Cyclopamine treatment led to a significant decrease in the percentage of HIOs with score 2 compared with control or SAG treatment, while SAG-treated HIOs had a significantly higher proportion of score 2 compared with controls (Figure 4F). Thus, HH signaling may be necessary to restrict WT1 localization to the outer cell layer in the developing intestine.

In order to determine whether HH modulation directly affects the intestinal serosa, we FACS isolated serosal mesothelial cells from suspension HIOs and treated these cells with SAG and cyclopamine. After 7 days, we used immunofluorescence to stain for WT1 and α SMA and calculated the percentage of cells in each condition that expressed each marker. We found that the percentage of α SMA⁺ cells in the cyclopamine group was significantly lower than the percentage of α SMA⁺ cells in both the basal media and SAG groups (Figure S5C), suggesting that HH inhibition may block the ability of mesothelial cells to differentiate into α SMA⁺ mesenchymal cells. This is consistent with observations in the mesothelium of the developing mouse lung (Dixit et al., 2013).

Inhibition of WNT signaling with IWR1 resulted in a majority of HIOs with a score of 1 (Figures 4G and S5B), suggesting that endogenous WNT signaling may be required for proper serosal mesothelial formation. Stimulation of WNT signaling with CHIR-99021 caused aberrant serosa formation and resulted in HIOs with score 0 or 3 (Figures 4G and S5B), suggesting that altered WNT signaling disrupts serosa formation. IWR1 treatment did not significantly alter HIO length, suggesting that WNT inhibition stunts mesothelial development without limiting the overall growth of suspension HIOs. Together, these results suggest that HH and WNT signaling are necessary for proper differentiation and patterning of the human intestinal serosal mesothelium.

DISCUSSION

In this work, we described suspension culture as an alternative to hydrogel or Matrigel culture for human intestinal organoids. Suspension culture provides an advantage over Matrigel by removing biological variability and reducing cost. Strikingly, HIOs cultured in non-adherent alginate or suspension formed a serosal mesothelium that resembled that of the human fetal intestine. scRNA-seq analysis of HIOs revealed a serosa population that was significantly similar to the human serosa cluster. We hypothesize that non-adherent culture conditions (alginate and suspension) promote mesothelial development compared with Matrigel, as HIO mesenchymal cells are able to self-organize rather than migrating away from the epithelium. There were some differences in gene expression between human and HIO serosal mesothelium, which can be attributed to differences between the *in vivo* environment and *in vitro* culture (Miller et al., 2020). Suspension culture enhances HIO serosa formation compared with alginate-grown HIOs, which suggests that lack of a compressive environment may enable mesothelial differentiation and organization. The emergence of a serosal mesothelium in suspension may be due to the fact that suspension culture mimics early developmental events *in vivo*, in which the gut tube is essentially suspended in fluid (Pansky, 1982).

In order to evaluate the functionality of HIO serosa, we devised a method to FACS purify mesothelial cells from suspension HIOs and confirmed that they undergo expected differentiation into smooth-muscle-like cells *in vitro*. Thus, HIOs are a promising model system to study mesothelial development and differentiation within a human model system *in vitro*. The ability to culture and sort mesothelial cells from HIOs may prompt further studies into factors driving mesothelial-to-mesenchymal differentiation. In addition, isolated

HIO serosa may be a promising source of mesothelial cells for therapeutic approaches (Colunga et al., 2019; Herrick and Mutsaers, 2007; Lachaud et al., 2015).

HIO serosa serves as an *in vitro* model system to study how the serosa originates from mesodermal progenitors in the developing human intestine. Contrary to reports on the mesothelium of the heart (Kruithof et al., 2006), we did not find mesothelial differentiation to be dependent on BMP or FGF. This may highlight organ- or species-specific differences in mesothelial development. We demonstrated that inhibition of WNT or HH signaling disrupted serosa formation in suspension HIOs. However, the precise mechanism by which WNT and HH play a role in serosal development, including whether these pathways have a direct versus indirect effect, is still unclear. FACS-isolated HIO serosal cells treated with cyclopamine do not have significantly increased WT1 expression but rather exhibit lower levels of α SMA expression compared with controls (Figure S5C), while whole HIOs treated with cyclopamine exhibit excess ectopic WT1 expression in the mesenchyme (Figures 4F and S5C). These data suggest that HH is not directly inducing or blocking serosa differentiation but may function to limit the ability of serosa to differentiate into other cell types, such as α SMA⁺ mesenchyme. Based on these results, we propose a model where blocking HH inhibits the ability of WT1⁺ serosal cells to differentiate into non-serosal cell types, such as smooth-muscle-like cells.

Limitations of the study

The yield of HIOs in suspension is reduced compared with Matrigel HIOs (Figure S1A), which could be due to heterogeneity in spheroids (Arora et al., 2017). While we have shown that HIO serosa is similar to human serosa, some mesothelial functions may be lost *in vitro*. Nonetheless, suspension HIOs present an improved method to study this poorly understood cell type. In addition, it is currently unclear how HH and WNT control serosa differentiation. HH and WNT inhibitors, activators, and ligands were added to the culture media of complete HIOs and thus likely impacted other cell types in the HIO in addition to the serosa. Moreover, combinatorial studies have yet to be carried out. To more precisely understand how these pathways work individually and in combination and how they influence a complex multi-tissue system like an HIO, activator and inhibitor experiments coupled with single-cell approaches may be needed. This work highlights the usefulness of suspension HIOs as a model to study human mesothelial development, but further work is necessary to determine specific mechanisms that control the development of this cell type.

STAR★METHODS

RESOURCE AVAILABILITY

Lead contact—Further information and requests for resources and reagents should be directed to and will be fulfilled by the Lead Contact, Dr. Jason Spence (spencejr@umich.edu).

Materials availability—This study did not generate new unique reagents.

Data and code availability—Sequencing data used in this study is deposited at EMBL-EBI ArrayExpress. Single-cell RNA sequencing of human tissue: human fetal intestine (ArrayExpress: E-MTAB-9489) (Holloway et al., 2021); human fetal intestine (ArrayExpress: E-MTAB-9363) (Holloway et al., 2020); human fetal intestine (ArrayExpress: E-MTAB-11335) – this study; HIO (ArrayExpress: E-MTAB-9228) (Holloway et al., 2020); HIO (ArrayExpress: E-MTAB-10187) (Yu et al., 2021); HIO (ArrayExpress: E-MTAB-10268) (Yu et al., 2021); HIO (ArrayExpress: E-MTAB-11338) – this study; HIO (ArrayExpress: E-MTAB-11347) – this study. Accession numbers for deposited data are also provided in the Key Resources Table. Code used for single cell analysis and data visualization can be found at: https://github.com/jason-spence-lab/Capeling_2022. Any additional information required to reanalyze the data reported in this work is available from the Lead Contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

hESC/hPSC lines and generation of hPSC-derived intestinal organoids—This study includes data from HIOs generated across 4 hPSC lines: Human ES lines H9 (NIH registry #0062, RRID: CVCL_9773, female) and UM63–1 (NIH registry #0277, RRID: CVCL_R782), as well as human iPSC lines WTC11 (RRID: CVCL_Y803, male) and 72.3 (McCracken et al., 2014). All experiments using hPSCs were approved by the University of Michigan Human Pluripotent Stem Cell Research Oversight Committee. hPSC lines and HIOs are routinely monitored for mycoplasma using the MycoAlert Mycoplasma Detection Kit (Lonza). hPSC lines are routinely karyotyped and H9 cells were authenticated using Short Tandem Repeat (STR) DNA profiling (Matsuo et al., 1999) at the University of Michigan DNA Sequencing Core and were found to exhibit an STR profile identical to previously described characteristics (Josephson et al., 2006).

Stem cell maintenance and differentiation into HIOs was carried out as recently described in detail (Capeling et al., 2019, 2020; Hill et al., 2017; McCracken et al., 2011; Tsai et al., 2017). Cell culture was carried out in a 37°C tissue culture incubator. hPSCs were maintained in mTeSR 1 or mTeSR Plus cultured media (Stemcell Technologies) and enzymatically passaged with dispase (Gibco). hPSCs underwent directed differentiation into definitive endoderm over a 3-day treatment with Activin A (100ng/mL, R & D Systems) added into RPMI media supplemented with 0%, 0.2%, 2% HyClone dFBS on subsequent days. Endoderm was differentiated into hindgut by treatment with FGF4 (500ng/mL (Sugawara et al., 2014)) and CHIR99021 (2μM, APEX-BIO).

Mid/Hindgut spheroids that budded off from the monolayer during differentiation were collected after days 5 and 6 of hindgut induction. Spheroids were embedded in alginate or Matrigel as previously described (Capeling et al., 2019), or transferred to low attachment plates for suspension culture to enable growth into HIOs. Organoids were maintained in basal growth media consisting of Advanced DMEM/F12 with 1X B27 (Thermo Fisher), GlutaMAX (Gibco, 1X), penicillin-streptomycin (Gibco, 100 U ml⁻¹ penicillin; 100 μg ml⁻¹ streptomycin), and HEPES buffer (Gibco, 15 mM). Organoid basal growth media was supplemented with epidermal growth factor (EGF) (R&D Systems; 100 ng/mL), Noggin-Fc (100ng/mL) (purified from conditioned media (Heijmans et al., 2013)), and R-Spondin1 (5%

conditioned medium (Ootani et al., 2009)) for the first three days of culture to promote patterning into proximal small intestine. On the third day after embedding, media was changed to basal growth media supplemented with EGF alone. HIOs were maintained in EGF-supplemented media for the duration of culture. Media was changed every 5–7 days. Organoids were not passaged to avoid disrupting the serosal mesothelium. Catalog information for all cell culture reagents described here can be found in the Key Resources Table.

Experiments involving pooled sets of HIOs involved $n \geq 10$ organoids per experiment. See Figure legends for more information on sample size for each experiment. For experimental treatment groups including the HIO inhibitor screen, equal numbers of HIOs were randomly allocated to each experimental group.

Human tissue—Normal, de-identified human fetal intestinal tissue was obtained from the University of Washington Laboratory of Developmental Biology and shipped overnight in Belzer-UW Cold Storage Solution (ThermoFisher, NC0952695) with cold packs as previously described (Miller et al., 2020). All research utilizing human tissue was approved by the University of Michigan institutional review board. For experiments involving human fetal small intestinal tissue, the following 8 samples were included: male, 47 days post-conception; female, 59 days post-conception; female, 72 days post-conception; male, 80 days post-conception; male, 85 days post-conception; male, 101 days post-conception; female, 127 days post-conception; female, 132 days post-conception.

METHOD DETAILS

Generation of low-attachment plates and suspension culture—Low attachment plates were generated using previously published methods (Choi et al., 2011). In summary, poly (2-hydroxyethyl methacrylate) (pHEMA) coating solution was prepared by dissolving 4g of pHEMA (Thermo Fisher, see Key Resources Table) into 40 mL of 95% ethanol with 10 mM NaOH. The solution was shaken immediately upon addition of pHEMA to avoid precipitation and rotated continuously overnight until fully dissolved. 1 mL of pHEMA coating solution was applied to each well of a 6 well plate inside a biosafety cabinet to generate a low attachment culture plate. The plate was rocked side to side to ensure distribution of pHEMA, and then pHEMA was collected for re-use. The low attachment plate was left in the biosafety cabinet overnight with the plate lid open to allow evaporation of excess coating solution, and UV was turned on for 15 minutes to sterilize the plate. Low attachment plates were rinsed twice with 1XPBS prior to addition of spheroids to remove excess pHEMA. Approximately 200 spheroids were transferred to 1 well of a 6-well low-attachment plate containing 5mL of media for suspension HIO culture. To change media without aspirating organoids in suspension, media was collected under a stereomicroscope using a P1000 after allowing organoids to settle to the bottom of the plate.

Embedding HIOs in alginate—Low-viscosity sodium alginate powder (Alfa Aesar) was dissolved in 1 mL of $1 \times$ PBS to a final concentration of 0.5%–2% (w/v) and heated to 98°C for 30 min on a heating block. Spheroids were suspended in alginate at a density of approximately 50 spheroids per 45 μ L. 5 μ L droplets of 2% (w/v) calcium chloride

(Sigma-Aldrich) were deposited on the bottom of 24-well tissue culture plates, and 45 μL of alginate containing spheroids was pipetted directly onto the calcium chloride solution to initiate ionic crosslinking. The gels polymerized at room temperature for 5–10 min and were then placed into a tissue culture incubator and allowed to fully set for 20 min at 37°C before media was added.

Mouse kidney capsule transplantation—The University of Michigan and Cincinnati Children’s Hospital Institutional Animal Care and Use Committees approved all animal research. HIOs were cultured for 4 weeks in suspension and then collected for transplantation, at which point HIOs were implanted under the kidney capsules of immunocompromised NOD-scid IL2Rg-null (NSG) mice (Jackson Laboratory strain no. 0005557) as previously described (Finkbeiner et al., 2015; Watson et al., 2014). In summary, mice were anaesthetized using 2% isoflurane. A left-flank incision was used to expose the kidney after shaving and sterilization with isopropyl alcohol. HIOs cultured in suspension were surgically implanted beneath mouse kidney capsules using forceps. Prior to closure, an intraperitoneal flush of Zosyn (100 mg kg^{-1} ; Pfizer) was administered. Mice were euthanized for retrieval of tHIOs after 8 weeks. Results shown are representative of one experiment performed with a total of $n=6$ mice, with at least one organoid implanted per kidney capsule depending on HIO size.

Flow cytometric analysis of HIOs—Suspension HIOs were transferred to a Petri dish containing TrypLE Express (Thermo Fisher) and mechanically cut into small pieces using a scalpel. TrypLE and dissociated HIOs were then transferred to a 15 mL conical tube and placed into a tissue culture incubator at 37°C. HIOs were digested in TrypLE at 37°C until the tissue was fully dissociated (~1.5 – 2 hours), and tissue was agitated roughly every 15 minutes during digestion by vortexing and pipetting up and down with a P1000. Once digestion was complete, reactions were quenched by adding a 2X volume of DMEM: F12 media. HIO suspensions were passed through a 70 μm filter and then centrifuged at 300g for 5 min at 4°C. Cells were then rinsed in staining buffer (1XPBS, 2% BSA, 1X PenStrep, 10 mM Y-27632 (Reagents Direct)), centrifuged, and re-suspended in an appropriate volume of staining buffer for antibody staining (100–200 μL). Cell suspensions were stained with conjugated FACS antibodies (Key Resources Table) and DAPI (0.2 $\mu\text{g}/\text{mL}$) at 4°C for 30 minutes. Cells were then rinsed with 3 mL of staining buffer, centrifuged, and re-suspended in 500 μL staining buffer. Flow cytometric analysis was performed using a Sony SY3200 cell sorter and accompanying software.

Cells were first gated on PDPN, and then passed through a secondary gate on ECAD and EPCAM. From the PDPN⁺ population, a PDPN⁺/ECAD⁺/EPCAM⁺-population was collected as HIO serosal mesothelium. From the PDPN⁻ population, a PDPN⁻/ECAD⁺/EPCAM⁺ population was collected as HIO epithelium, and a PDPN⁻/ECAD⁻/EPCAM⁻-population was collected as HIO mesenchyme.

Culture of isolated HIO-serosa—Following FACS isolation, PDPN⁺/ECAD⁺/EPCAM⁺-serosa was plated on Matrigel-coated 24-well plates in basal organoid growth media supplemented with 10 mM Y-27632 (Reagents Direct) and 0.4 $\mu\text{g}/\text{mL}$ Hydrocortisone (Sigma-Aldrich) (Yung et al., 2006). Y-27632 was removed after 24 hours for 7-day cultures.

For HH modulation experiments, Cyclopamine (5 μ M) or SAG (2 μ M) was added to basal organoid growth media with 0.4 μ g/mL Hydrocortisone for 6 days following the 24-hour culture in basal media. See Key Resources Table.

HIO-serosa signaling screen—Hindgut spheroids were collected and cultured in suspension with basal organoid growth medium supplemented with (EGF) (100 ng/mL), Noggin-Fc (100ng/mL), and R-Spondin2 (5% conditioned medium) for 3 days. On the third day after collection, media was changed to basal organoid growth medium supplemented with only EGF (100 ng/mL). After 7 days, signaling activators or inhibitors were applied by changing cultured medium to organoid growth medium supplemented with EGF (100 ng/mL – basal medium control condition) and signaling compounds, including DAPT (10 μ M ng/mL), IWR1 (10 μ M), CHIR99021 (2 μ M), Cyclopamine (5 μ M), SAG (2 μ M), SU5402 (10 μ M), Noggin-Fc (100 ng/mL), BMP4 (100 ng/mL), SHH (33.3 ng/mL), and WNT3A (250 ng/mL). See Key Resources Table for catalog information. HIOs were then cultured in medium containing signaling compounds for 3 weeks (28 days total in culture). After 28 days, HIOs were collected and fixed for immunofluorescence staining or flash-frozen for qRT-PCR analysis.

To score HIOs and determine effects of signaling compounds on serosal mesothelium, an entire batch of matched HIOs was fixed and stained for WT1 and pCK. Each HIO in one plane of section was given a score between 0 and 3 based on WT1 and pCK immunofluorescent staining. Scoring was as follows – 0: no WT1+/pCK+ serosal mesothelium present or aberrant WT1 expression in the mesenchyme but not outer layer. 1: partial WT1+/pCK+ serosal mesothelium on the outside of the HIO but not covering the entire organoid. 2: WT1+/pCK+ serosal mesothelium covering the entire HIO. 3: complete or partial WT1+/pCK+ serosal mesothelium on the outside of the HIO combined with ectopic WT1 expression throughout the mesenchyme.

Single cell preparation of tissue for single cell RNA sequencing—Previously published methods were utilized to carry out cell dissociations (Miller et al., 2020). Tubes and pipette tips were washed with HBSS containing 1% BSA prior to dissociation protocol to prevent adhesion of cell suspensions to the plastic. Centrifugation steps were carried out at 10°C unless otherwise stated. The Neural Tissue Dissociation Kit (Miltenyi, cat. no. 130–092-628) was used for all RNA-seq dissociations.

Human fetal intestinal tissue or human intestinal organoids was dissociated into single cells by first mechanically cutting tissue into small pieces using a scalpel in a Petri dish containing cold 1X HBSS with Mg^{2+} , Ca^{2+} . Tissue fragments were then transferred to a 15 mL conical tube and treated with Mix 1 for 15 minutes 10°C. Mix 2 was added to the tube and the remainder of the digestion was carried out at 10°C until tissue was fully digested. Every 10 minutes, the cell solution was pipetted up and down with a P1000 and assessed under a stereo microscope to determine whether the digestion was complete. Once tissue was digested into a single-cell suspension, cells were passed through a 70 μ m filter coated with 1% BSA in 1X HBSS. The filtered cell suspension was centrifuged for 5 minutes at 500g, and the cell pellet was resuspended in 500 μ L 1X HBSS with Mg^{2+} , Ca^{2+} . 1 mL Red Blood Cell Lysis buffer (Roche cat. No 11814389001) was added to the cell suspension and

the tube containing cells was placed on a rocker at 4°C for 15 minutes. The cell suspension was centrifuged for 5 minutes at 500g, and the pellet was washed by resuspending in 2 mL HBSS with 1% BSA. Centrifugation and re-suspension were repeated such that cells were washed twice and then counted with a hemocytometer. Cells were then centrifuged and resuspended to reach a final concentration of 1000 cells/μL and kept on ice while transferred to the University of Michigan Advanced Genomics Core where single cell droplets were immediately prepared on the 10x Chromium according to manufacturer instructions. A target of 5,000–10,000 cells captured was utilized. The Chromium Next GEM Single Cell 3' Library Construction Kit v3.1 (10x Genomics) was used to prepare single cell libraries according to manufacturer instructions.

RNA extraction and quantitative RT-PCR analysis—qRT-PCR experiments were carried out as previously described (Miller et al., 2018). RNA was extracted using the MagMAX-96 Total RNA Isolation System (Life Technologies) or Qiagen RNeasy Mini kit for FACS samples. A Nanodrop 2000 spectrophotometer (Thermo Scientific) was used to assess RNA quality and concentration, and then a cDNA library was generated from isolated RNA using the SuperScript VILO cDNA master mix kit (Invitrogen). qRT-PCR analysis was conducted using the QuantiTect SYBR Green PCR Kit (Qiagen) on a Step One Plus Real-Time PCR system (Life Technologies). Gene expression levels were calculated as a change relative to GAPDH or ECAD expression using arbitrary units, which were calculated by the following equation: $2^{-(\text{GAPDH/ECAD Ct} - \text{Gene Ct})} \times 10,000$. Expression was normalized to ECAD for epithelial genes as there were variable levels of epithelium between samples. A Ct value of 40 or greater was considered not detectable. A list of primer sequences used can be found in the Key Resources Table and Table S2.

Tissue preparation, immunohistochemistry, and imaging

Paraffin sectioning and staining: HIO and tHIO tissues were fixed in 4% Paraformaldehyde (Sigma) overnight, washed with PBS, and then dehydrated in an alcohol series: 30 minutes each in 25%, 50%, 75% Methanol:PBS/0.05% Tween-20, followed by 100% Methanol, 100% Ethanol and 70% Ethanol. Tissue was processed into paraffin using an automated tissue processor (Leica ASP300). Paraffin blocks were sectioned 7 μm thick, and immunohistochemical staining was performed as previously described (Spence et al., 2009). Briefly, slides were rehydrated in a series of HistoClear, 100% Ethanol, 95% Ethanol, 70% Ethanol, 30% Ethanol, DI H₂O with 2 changes of 3 minutes each. Antigen retrieval as performed in 1X sodium citrate buffer in a vegetable steamer for 40 minutes. Following antigen retrieval, slides were washed in PBS and permeabilized for 10 minutes in 0.1% TritonX-100 in 1xPBS, blocked for 45 minutes in 0.1% Tween-20, 5% normal donkey serum in 1XPBS. Antibodies used in this study can be found in the Key Resources Table. Primary antibodies were diluted in block and applied overnight at 4°C. Slides were then washed 3 times in 1X PBS. Secondary antibodies and DAPI were diluted in block and applied for 40 minutes at room temperature. Slides were then washed 3 times in 1X PBS and cover slipped with ProLong Gold.

H&E staining was performed using Harris Modified Hematoxylin (FisherScientific) and Shandon Eosin Y (ThermoScientific) according to manufacturer's instructions. Alcian

blue/PAS staining was performed using the Newcomer supply Alcian Blue/PAS Stain kit (Newcomer Supply, Inc.) according to manufacturer's instructions. Trichrome staining was performed by the University of Michigan *in vivo* Animal Core.

Imaging and image processing: Fluorescently stained slides were imaged on a Nikon A-1 confocal microscope. Brightness and contrast adjustments were carried out using ImageJ (National Institute of Health, USA) (Schneider et al., 2012) and adjustments were made uniformly across images.

Transmission electron microscopy: Suspension HIOs cultured for 28 days were collected for TEM and prepared using conventional TEM sample preparation methods described by the University of Michigan BRCF Microscopy and Image Analysis Laboratory. HIOs were fixed in 3% glutaraldehyde + 3% paraformaldehyde in 0.1M cacodylate buffer (CB), pH 7.2 until ready for sample prep. Samples were then washed 3 times for 15 minutes in 0.1M CB. After washing, samples were processed for 1 hour on ice in a post-fixation solution of 1.5% $K_4Fe(CN)_6$ + 2% OsO_4 in 0.1M CB. Samples were then washed 3 times in 0.1M CB, and 3 times in 0.1M Na_2 + Acetate Buffer, pH 5.2, followed by *en bloc* staining for 1 hour in 2% Uranyl Acetate + 0.1M Na_2 + Acetate Buffer, pH 5.2. Samples were then processed overnight in an automated tissue processor, including dehydration from H_2O through 30%, 50%, 70%, 80%, 90%, 95%, 100% ethanol, followed by 100% acetone. Samples were infiltrated with Spurr's resin at a ratio of acetone: Spurr's resin of 2:1 for 1 hour, 1:1 for 2 hours, 1:2 for 16 hours, and absolute Spurr's resin for 24 hours. After embedding and polymerization, samples were sectioned on an ultramicrotome. TEM sample grids were imaged on a JEOL JEM 1400 PLUS TEM.

ELISA for t-PA expression: HIOs were cultured in Matrigel or suspension for 28 days *in vitro*. Conditioned media was collected from $n > 3$ wells of HIOs after 28 days and stored at $-80^\circ C$ until testing. Expression of human t-PA in the conditioned media was detected using a human t-PA ELISA kit (ThermoFisher) according to manufacturer's instructions. The absorbance of each microwell was read on a Molecular Devices SpectraMax M5e microplate reader, using 450 nm as the primary wave length.

QUANTIFICATION AND STATISTICAL ANALYSIS

Statistical analyses and plots were generated in Prism 8 software (GraphPad). If more than two groups were being compared within a single experiment, an unpaired one-way ANOVA was performed followed by Tukey's multiple comparisons test to compare the mean of each group with the mean of every other group within the experiment unless otherwise specified. For all statistical tests, a significance value of 0.05 was used. For every analysis, the strength of p values is reported in the figures according the following: $p > 0.05$, $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$, $****p \leq 0.0001$. Details of statistical tests can be found in the figure legends. With the exception of scRNA-seq, three hPSC lines were used across experiments with at least 3 independent experiments and at least 3 technical replicates per experiment.

Computational analysis of single-cell RNA sequencing data

Overview: To visualize distinct cell populations within the single-cell RNA sequencing dataset, we employed the general workflow outlined by the Scanpy Python package (Wolf et al., 2018). This pipeline includes the following steps: filtering cells for quality control, log normalization of counts per cell, extraction of highly variable genes, regressing out specified variables, scaling, reducing dimensionality with principal component analysis (PCA) and uniform manifold approximation and projection (UMAP) (Becht et al., 2019; McInnes et al., 2018), and clustering by the Louvain algorithm (Blondel et al., 2008).

Sequencing data and processing FASTQ reads into gene expression matrices: All single-cell RNA sequencing was performed at the University of Michigan Advanced Genomics Core with an Illumina Novaseq 6000. The 10x Genomics Cell Ranger pipeline was used to process raw Illumina base calls (BCLs) into gene expression matrices. BCL files were demultiplexed to trim adaptor sequences and unique molecular identifiers (UMIs) from reads. Each sample was then aligned to the human reference genome (hg19) to create a filtered feature bar code matrix that contains only the detectable genes for each sample.

Quality control: To ensure quality of the data, all samples were filtered to remove cells expressing too few or too many genes (Figure S2C-S2E/Figure 2A - <500 , >10000 ; Figures 2B and 2C/S2F - <500 , >8000 ; Figures 3/S3 - <500 , >8000 ; Figures S3F and S3G/4A - <500 , >10000), with high UMI counts (Figures S2C-S2E/2A - 60000; Figures 2B and 2C/S2F - 50000; Figures 3/S3 - 50000; Figures S3F and S3G/4A - 60000), or a fraction of mitochondrial genes greater than 0.1.

Normalization and scaling: Data matrix read counts per cell were log normalized, and highly variable genes were extracted. Using Scanpy's simple linear regression functionality, the effects of total reads per cell and mitochondrial transcript fraction were removed. The output was then scaled by a z-transformation. Following these steps, a total of (Figure S2C-S2E/2A - 51790 cells, 1710 genes; Figures 2B and 2C/S2F - 4619 cells, 4167 genes; Figures 3/S3 - 13055 cells, 2117 genes; Figures S3F and S3G/4A - 52689 cells, 2615 genes) were kept for clustering and visualization.

Variable gene selection: Highly variable genes were selected by splitting genes into 20 equal-width bins based on log normalized mean expression. Normalized variance-to-mean dispersion values were calculated for each bin. Genes with log normalized mean expression levels between 0.125 and 3 and normalized dispersion values above 0.5 were considered highly variable and extracted for downstream analysis.

Batch correction: We have noticed batch effects when clustering data due to technical artifacts such as timing of data acquisition or differences in dissociation protocol. To mitigate these effects, we used the Python package BBKNN (batch balanced k nearest neighbors) (Polański et al., 2020). BBKNN was selected over other batch correction algorithms due to its compatibility with Scanpy and optimal scaling with large datasets. This tool was used in place of Scanpy's nearest neighbor embedding functionality. BBKNN uses a modified procedure to the k nearest neighbors' algorithm by first splitting the dataset

into batches defined by technical artifacts. For each cell, the nearest neighbors are then computed independently per batch rather than finding the nearest neighbors for each cell in the entire dataset. This helps to form connections between similar cells in different batches without altering the PCA space. After completion of batch correction, cell clustering should no longer be driven by technical artifacts.

Dimension reduction and clustering: Principal component analysis (PCA) was conducted on the filtered expression matrix followed. Using the top principal components, a neighborhood graph was calculated for the nearest neighbors (Figures S2C-S2E/2A – 16 principal components, 30 neighbors; Figures 2B and 2C/S2F – 12 principal components, 15 neighbors; Figures 3/S3 – 12 principal components, 15 neighbors; Figures S3F and S3G/4A – 16 principal components, 30 neighbors). BBKNN was implemented when necessary and calculated using the top 50 principal components with 3 neighbors per batch. The UMAP algorithm was then applied for visualization on 2 dimensions. Using the Louvain algorithm, clusters were identified with a resolution of (Figures S2C-E/2A – 0.8; Figures 2B and 2C/S2F – 0.35; Figures 3/S3 – 0.65, Figures S3F and S3G/4A - 1).

Cluster annotation: Using canonically expressed gene markers, each cluster's general cell identity was annotated. Markers utilized include epithelium (*CDH1*, *EPCAM*, *CDX2*, *PDX1*, *VIL1*, *CLDN4*), mesenchyme (*VIM*, *COL1A2*, *PDGFRA*, *DCN*, *TCF21*, *COL3A1*, *FOXF1*), neuronal (*POSTN*, *S100B*, *STMN2*, *ELAV4*), endothelial (*ESAM*, *CDH5*, *CD34*, *KDR*), immune (*CD53*, *VAMP8*, *CD48*, *ITGB2*), and serosal mesothelium (*MSLN*, *WT1*, *UPK3B*, *PDPN*).

Sub-clustering: After annotating clusters within the UMAP embedding, specific clusters of interest were identified for further sub-clustering and analysis. The corresponding cells were extracted from the original filtered but unnormalized data matrix to include (Figure 2A – 1040 cells, Figure 2C – 962 cells, Figure 3B and 3C – 1903 cells, Figure S3G, Figure 4A - 8179). The extracted cell matrix then underwent log normalization, variable gene extraction, linear regression, z transformation, and dimension reduction to obtain a 2-dimensional UMAP embedding for visualization.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Highlights

- Suspension culture supports growth of hPSC-derived human intestinal organoids (HIOs)
- Suspension HIOs form an outer serosal mesothelial layer
- HIO serosa is molecularly and functionally similar to human intestinal serosa
- HH and WNT signaling play a role in differentiation and patterning of the serosa

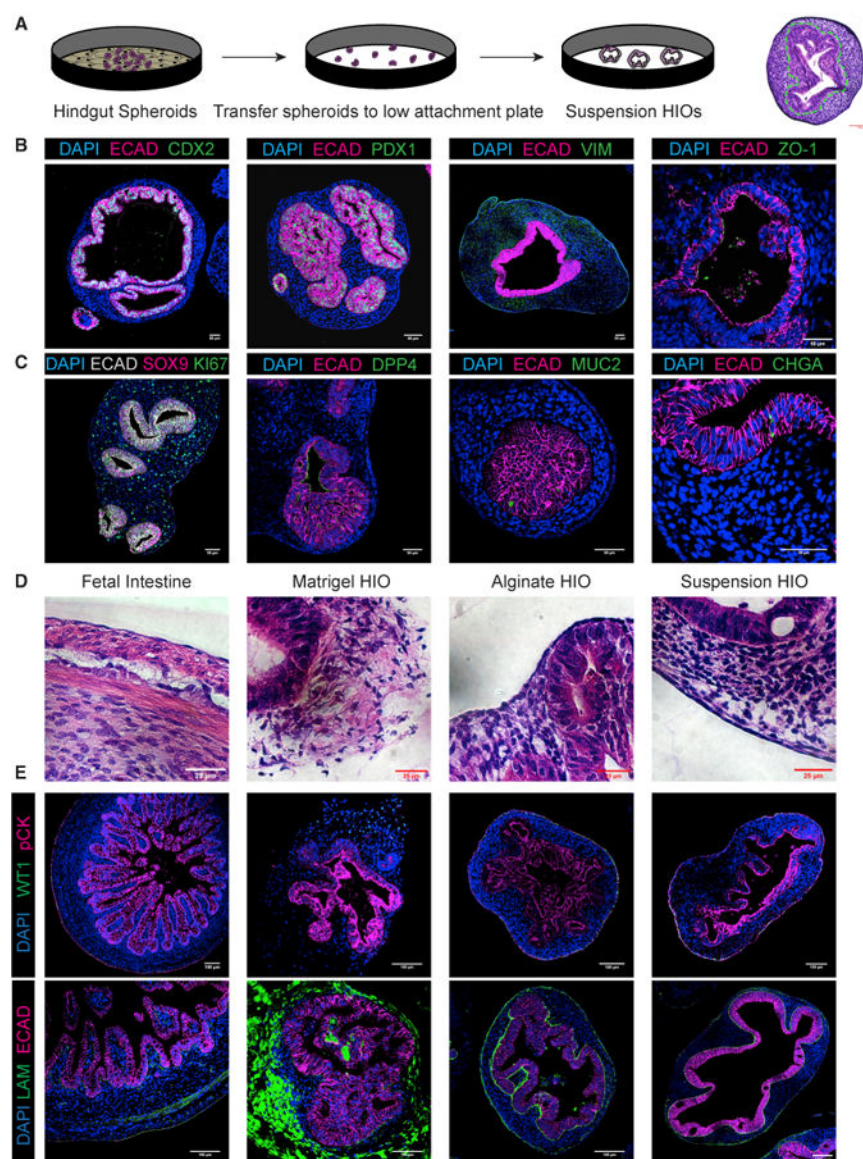


Figure 1. Suspension culture supports HIO growth and serosa formation

(A) Schematic depicting HIO suspension culture. Right: hematoxylin and eosin (H&E) stain of a 28-day suspension HIO is shown. The epithelium is outlined in green.

(B) Representative images of HIOs cultured in suspension for 28 days. Markers shown are ECAD, CDX2, PDX1, VIM, and ZO-1. Scale bars represent 50 μm .

(C) Representative images of epithelial cell marker staining in HIOs cultured in suspension for 28 days. Markers shown are SOX9, KI67, DPP4, MUC2, and CHGA. Scale bars represent 50 μm .

(D) H&E staining of the outer mesenchymal layer in human fetal intestine compared with HIOs cultured in Matrigel, 1% alginate, and suspension for 28 days. Scale bars represent 25 μm .

(E) Representative images of mesothelial markers in human fetal intestine compared with HIOs cultured in Matrigel, 1% alginate, and suspension for 28 days. Markers shown are WT1, pan-Cytokeratin (pCK), LAM, and ECAD. Scale bars represent 100 μm .

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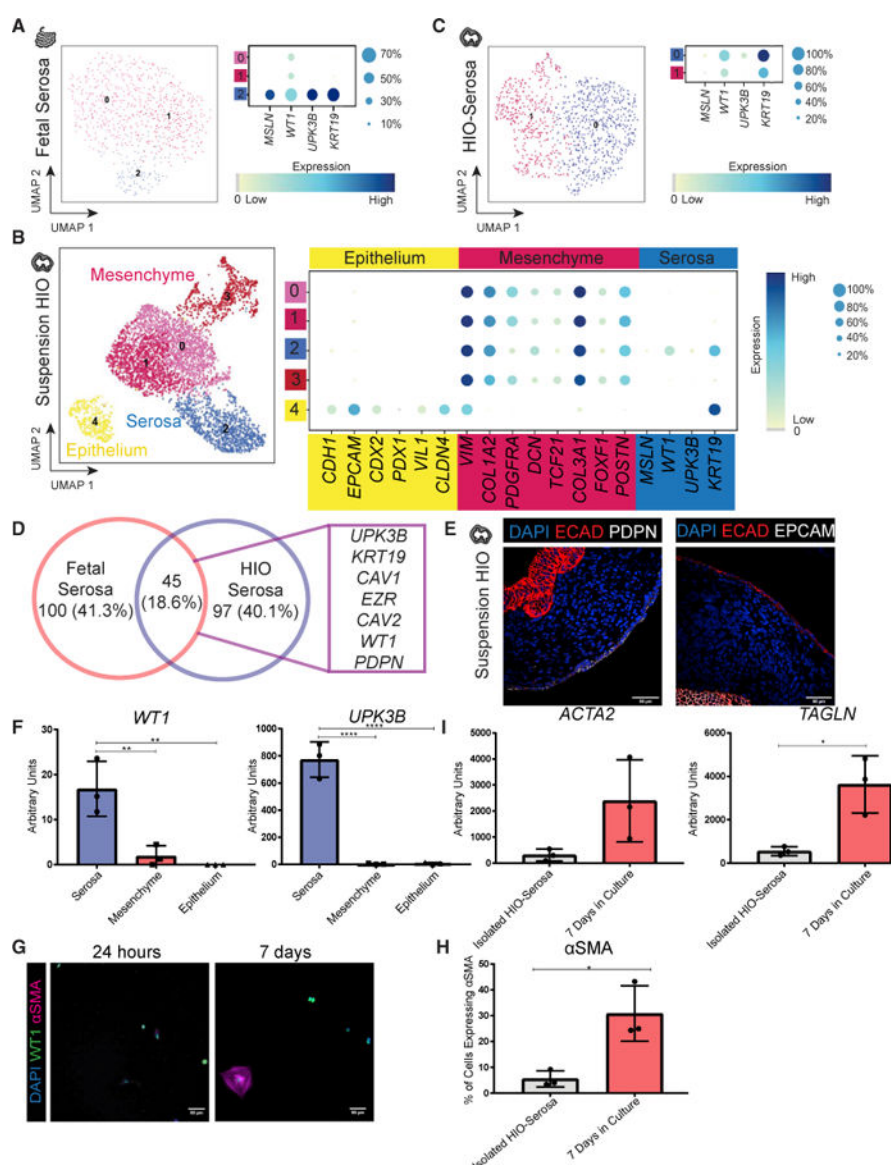


Figure 2. HIO serosa is molecularly and functionally similar to human serosa

(A) Left: UMAP plot of 761 cells computationally extracted from a sub-cluster within cluster 8 of the analysis on human fetal small intestine (Figure S2). Right: dot plot of mesothelial genes highlighting the mesothelial population within cluster 2 (human fetal serosa) is shown.

(B) Left: UMAP plot of 4,619 cells from 28-day suspension HIOs. Right: dot plot of genes in suspension HIOs associated with major cell classes is shown.

(C) Left: UMAP plot of 989 cells computationally extracted from cluster 2 of the analysis on suspension HIOs (B). Right: dot plot of mesothelial genes highlighting the mesothelial population within cluster 0 (HIO serosa) is shown.

(D) Venn diagram depicting genes that are significantly enriched ($\log_2$ fold >1.5 in expression of a gene in the serosa cluster relative to all other clusters) in human fetal serosa (cluster 2; A) and suspension HIO serosa (cluster 0; C).

- (E) Representative images of mesothelial markers used for FACS sorting in 28-day suspension HIOs (ECAD, EPCAM, and PDPN). Scale bars represent 50 μm .
- (F) qRT-PCR analysis of *WT1* and *UPK3B* expression in FACS-isolated serosa, mesenchyme, and epithelium from suspension HIOs. * $p \leq 0.05$, ** $p \leq 0.01$, *** $p \leq 0.001$, **** $p \leq 0.0001$.
- (G) Representative images of FACS-sorted HIO serosa stained for WT1 and αSMA after 24 h or 7 days cultured *in vitro*. Scale bars represent 50 μm .
- (H) Quantification of the percentage of DAPI⁺ FACS-sorted HIO serosa cells that express αSMA . Each point represents one independent experiment with cells isolated from suspension HIOs from three different hPSC lines. * $p \leq 0.05$.
- (I) qRT-PCR analysis of *ACTA2* and *TAGLN* expression in FACS-sorted HIO serosa cells immediately after sorting or after 7 days in culture. * $p \leq 0.05$. For all graphs, data represent the mean $\pm$ standard error of the mean.
- For all dot plots, dot size denotes the proportion of cells in a cluster expressing each marker and dot color indicates expression level within the cluster based on normalized *Z* score. Significance was calculated with a one-way ANOVA and multiple comparisons test. For all qRT-PCR data, expression levels are normalized to GAPDH. Each point is representative of sorting results from one pooled batch of HIOs, across $n = 3$ independent experiments (biological replicates).

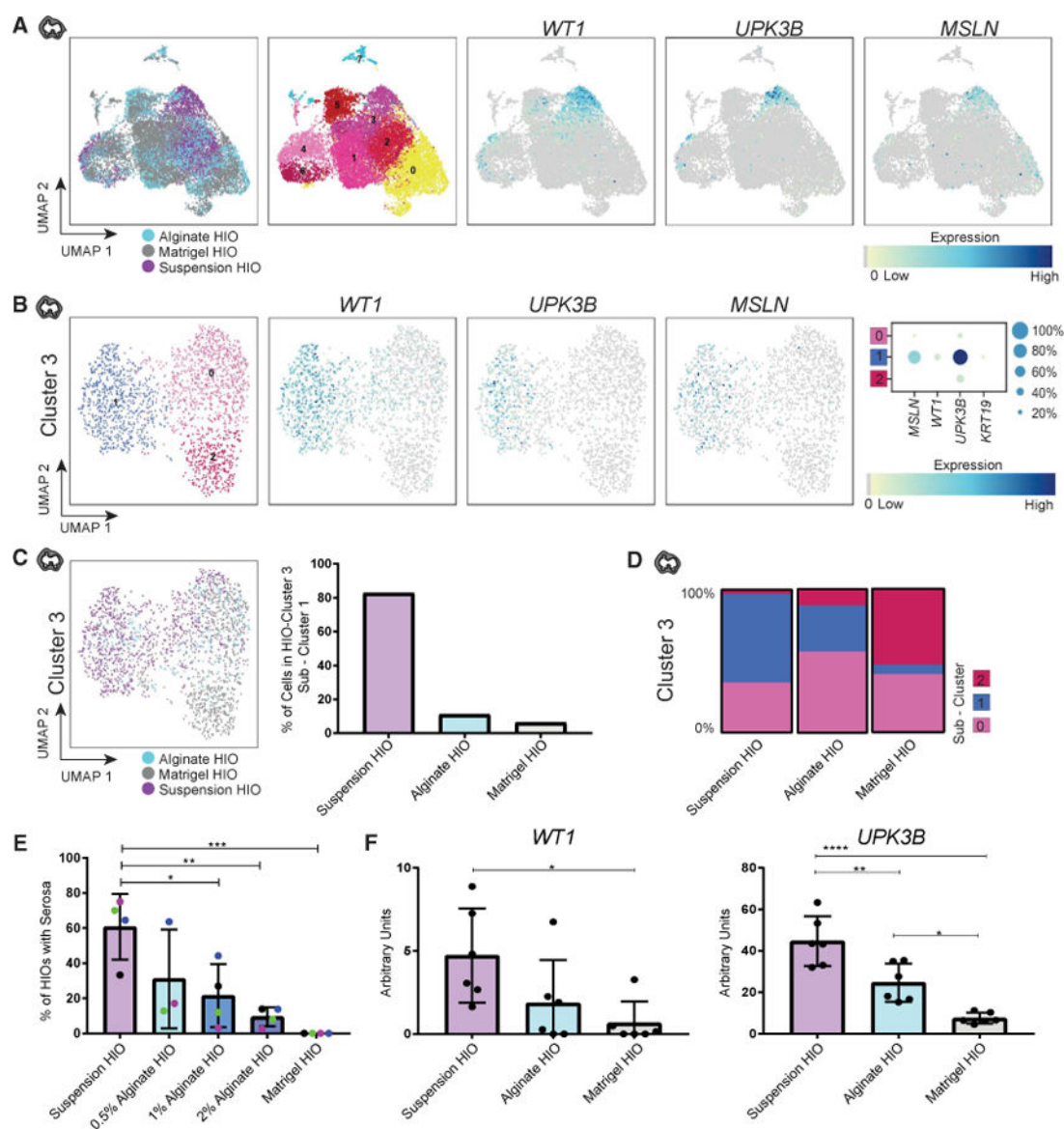


Figure 3. Comparing HIO serosa between Matrigel, alginate, and suspension culture

(A) UMAP plot of 13,055 cells from 28-day alginate, Matrigel, and suspension HIOs. The

batch correction algorithm BBKNN was utilized. Feature plots for mesothelial markers

WT1, *MSLN*, and *UPK3B* highlight the mesothelial population within cluster 3.

(B) Left: UMAP plot of 1,903 cells computationally extracted from cluster 3 of the analysis

on alginate, Matrigel, and suspension HIOs (A). Middle: feature plots for mesothelial

markers *WT1*, *MSLN*, and *UPK3B* highlight the mesothelial population within cluster 1.

Right: dot plot of mesothelial genes is shown.

(C) Left: UMAP plot of computationally extracted HIO serosa (cluster 3, sub-cluster 1)

separated by culture condition. Right: quantification of the percentage of cells within cluster

3, sub-cluster 1 that are derived from each condition is shown.

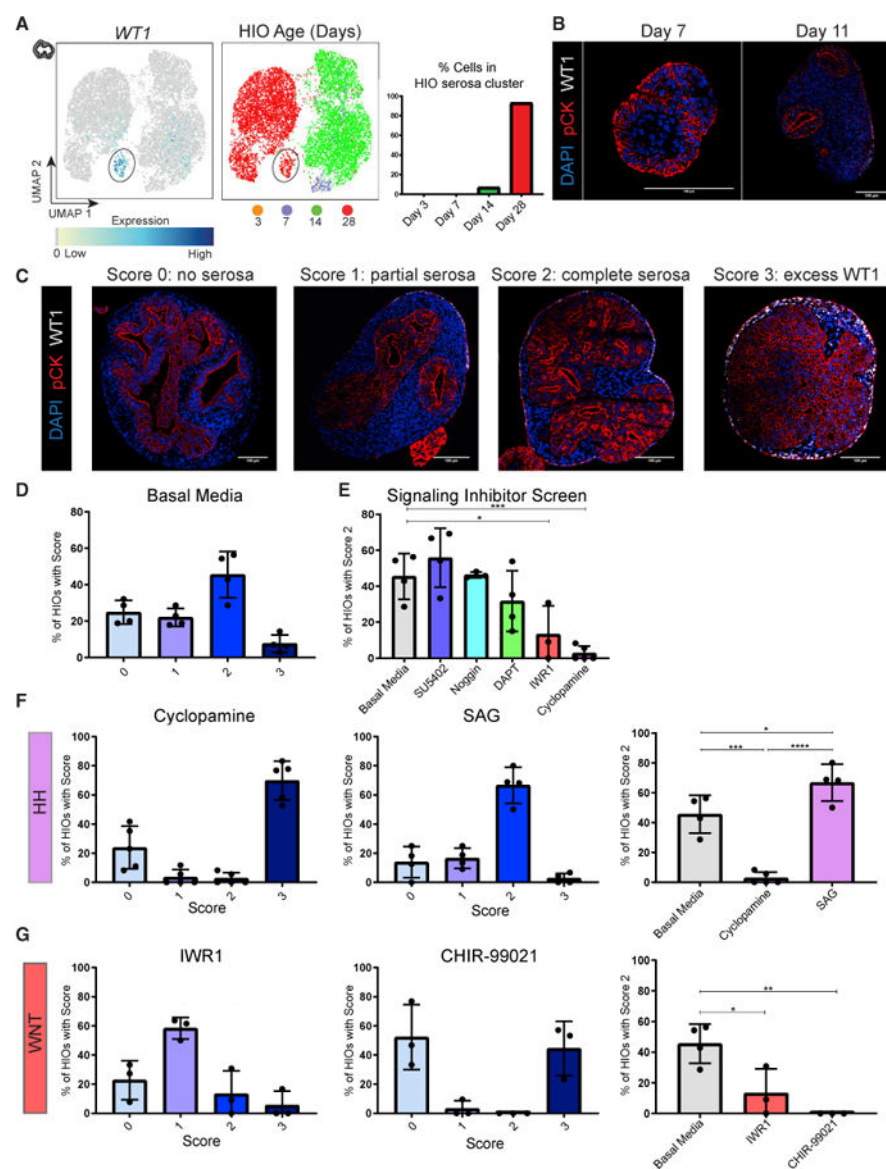
(D) Distribution of the cells from alginate, Matrigel, and suspension HIOs in cluster 3 that

were extracted into sub-clusters 0, 1, or 2.

(E) Quantification of the percentage of HIOs in four independent batches that form a full or partial WT1⁺ pCK⁺ outer serosa. Each dot represents data from one batch, each color dot depicts an independent batch of HIOs (biological replicates), and bars depict mean and standard error. Significance was calculated with a one-way ANOVA and multiple comparisons test. *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001.

(F) qRT-PCR analysis of *WT1* and *UPK3B* expression in HIOs derived from two independent hPSC lines cultured in suspension for 28 days *in vitro*. Expression levels are normalized to GAPDH. Each point is representative of 6–10 HIOs pooled from the same batch (biological replicates).

Data represent the mean ± standard error of the mean. Significance was calculated with a one-way ANOVA and multiple comparisons test. *p ≤ 0.05, **p ≤ 0.01, ***p ≤ 0.001, ****p ≤ 0.0001.



IWR1 (WNT), and cyclopamine (HH). Significance was calculated in relation to the control basal media condition. $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$.

(F) Quantification of the percentage of HIOs from four independent experiments that scored 0–3 in basal media or media supplemented with SAG or cyclopamine. Right: comparison of the percentage of HIOs that received score 2 in basal media with SAG and cyclopamine treatment is shown. $*p \leq 0.05$, $**p \leq 0.01$, $***p \leq 0.001$, $****p \leq 0.0001$.

(G) Quantification of the percentage of HIOs from three independent experiments that scored 0–3 in basal media or media supplemented with IWR1 or CHIR-99021. Right: comparison of the percentage of HIOs that received score 2 in basal media with IWR1 and CHIR-99021 treatment is shown. $*p \leq 0.05$, $**p \leq 0.01$.

For all graphs, data represent the mean $\pm$ standard error of the mean. Significance was calculated with a one-way ANOVA and multiple comparisons test.

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Goat polyclonal anti-E-cadherin	R&D Systems	Cat#AF748; RRID: AB_355568
Mouse monoclonal anti-E-cadherin	BD Transduction Laboratories	Cat#610181; RRID: AB_397580
Goat monoclonal anti-Vimentin	R&D Systems	Cat#MAB2105; RRID: AB_2241653
Mouse monoclonal anti-CDX2	BioGenex	Cat#MU392A-UC; RRID: AB_2650531
Rabbit monoclonal anti-PDX1	Cell Signaling Technology	Cat#5679; RRID: AB_10706174
Rabbit monoclonal anti-ZO-1	Cell Signaling Technology	Cat# 13663; RRID: AB_2798287
Rabbit polyclonal anti-MUC2	Santa Cruz Biotechnology	Cat# sc-15334; RRID: AB_2146667
Goat polyclonal anti-CHGA	Santa Cruz Biotechnology	Cat# sc-1488; RRID: AB_2276319
Goat polyclonal anti-SOX9	R&D Systems	Cat# AF3075; RRID: AB_2194160
Rabbit monoclonal anti-KI67	Thermo Scientific	Cat# RM-9106-S1; RRID: AB_149792
Goat polyclonal anti-DPPIV	R&D Systems	Cat# AF954; RRID: AB_355739
Goat polyclonal anti-LYZ	Santa Cruz Biotechnology	Cat# sc-27958; RRID: AB_2138790
Rabbit monoclonal anti-DEFA5	Abcam	Cat# ab180515; RRID: AB_2864387
Rabbit polyclonal anti-OLFM4	Abcam	Cat# ab85046; RRID: AB_10670544
Mouse monoclonal anti- α SMA-Cy3	Sigma-Aldrich	Cat# C6198; RRID: AB_476856
Rabbit monoclonal anti-WT1	Abcam	Cat# ab89901; RRID: AB_2043201
Mouse monoclonal anti-pan Cytokeratin	Abcam	Cat# ab86734; RRID: AB_10674321
Rabbit polyclonal-anti Laminin	Abcam	Cat# ab11575; RRID: AB_298179
Rabbit polyclonal anti-EpCAM	Sigma-Aldrich	Cat# HPA026761; RRID: AB_1848198
Rabbit polyclonal anti-PDPN	Santa Cruz Biotechnology	Cat# sc-134482; RRID: AB_2162079
Rabbit polyclonal anti-CAV1	Atlas Antibodies	Cat# HPA049326; RRID: AB_2680714
Rabbit polyclonal anti-CAV2	Atlas Antibodies	Cat# HPA044810; RRID: AB_2679098
Mouse monoclonal anti-Ezrin	Sigma-Aldrich	Cat# E8897; RRID: AB_476955
Goat polyclonal anti-GATA4	Santa Cruz Biotechnology	Cat# sc-1237; RRID: AB_2108747
Monoclonal anti-ECAD-PE	Miltenyi Biotec	Cat# 130-111-992; RRID: AB_2657482
Monoclonal anti-EpCAM-FITC	Miltenyi Biotec	Cat# 130-111-115; RRID: AB_2657492
Monoclonal anti-PDPN-APC	Miltenyi Biotec	Cat# 130-107-016; RRID: AB_2653263
Donkey anti-mouse 488	Jackson ImmunoResearch	Cat#715-545-150; RRID: AB_2340846
Donkey anti-mouse Cy3	Jackson ImmunoResearch	Cat#715-165-150; RRID: AB_2340813
Donkey anti-mouse 647	Jackson ImmunoResearch	Cat#715-605-150; RRID: AB_2340862
Donkey anti-rabbit 488	Jackson ImmunoResearch	Cat#711-545-152; RRID: AB_2313584
Donkey anti-rabbit Cy3	Jackson ImmunoResearch	Cat#711-165-152; RRID: AB_2307443
Donkey anti-rabbit 647	Jackson ImmunoResearch	Cat#711-605-152; RRID: AB_2492288
Donkey anti-goat 488	Jackson ImmunoResearch	Cat#705-545-147; RRID: AB_2336933
Donkey anti-goat Cy3	Jackson ImmunoResearch	Cat#705-165-147; RRID: AB_2307351
Donkey anti-goat 647	Jackson ImmunoResearch	Cat#705-605-147; RRID: AB_2340437

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Human Fetal Small Intestine	University of Washington Laboratory of Developmental Biology	N/A
Chemicals, peptides, and recombinant proteins		
Activin A	R&D	Cat#338-AC
FGF4	Purified in house	(Sugawara et al., 2014)
CHIR99021	APExBIO	Cat# A3011
Epidermal Growth Factor (EGF)	R&D Systems	Cat# 236-EG
Recombinant Human Noggin-FC, purified from HEK293 cells expressing FC-tagged Noggin	Purified in house	(Heijmans et al., 2013)
Human R-Spondin1 Conditioned Medium from Cultrex HA-R- Spondin1-Fc 293 T Cells	R&D Systems	Cat# 3710–001-01
B27 supplement	Thermo Fisher	Cat#17504044
HEPES	Thermo Fisher	Cat#15630080
GlutaMAX	Gibco	Cat#35050061
Penicillin-Streptomycin	Gibco	Cat# 15070063
Dispase	Life Technologies	Cat#17105–041
Poly (2-hydroxyethyl methacrylate) (pHEMA)	Sigma-Aldrich	Cat#P3932
Calcium Chloride	Sigma-Aldrich	Cat# 449709
TrypLE Express Enzyme	Gibco	Cat# 12604013
Hydrocortisone	STEMCELL Technologies	Cat#07904
Y27632	Stemgent	Cat#04–0012
SAG	STEMCELL Technologies	Cat#73414
Cyclopamine	Selleckchem	Cat#S1146
IWR1	R&D Systems	Cat#3532
SU5402	Santa Cruz Biotechnology	Cat#215543–92-3
DAPT	Calbiochem	Cat#565784
SHH	R&D Systems	Cat#1845-SH
WNT3A	R&D Systems	Cat# 5036-WN
Red Blood Cell Lysis Buffer	Roche	Cat#11814389001
Critical commercial assays		
Neural Tissue Dissociation Kit (P)	Miltenyi	Cat#130–092-628
SuperScript VILO cDNA Synthesis Kit	ThermoFisher	Cat#11754250
MagMAX-96 Total RNA Isolation Kit	Ambion	Cat#AM1830
QuantiTect SYBR Green PCR Kit	Qiagen	Cat#204145
Chromium Next GEM Single Cell 3' Library Construction Kit v3	10x Genomics	Cat#PN-1000075
Chromium Next GEM Single Cell 3' Library Construction Kit v2	10x Genomics	Cat#PN-120237
Human t-PA ELISA Kit	ThermoFisher	Cat#BMS258–2
Deposited data		
Raw scRNA-seq data (human fetal duodenum)	Holloway et al. (2021)	ArrayExpress: E-MTAB-9489
Raw scRNA-seq data (human fetal small intestine)	Holloway et al. (2020)	ArrayExpress: E-MTAB-9363
Raw scRNA-seq data (human fetal small intestine)	This study	ArrayExpress: E-MTAB-11335

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Raw scRNA-seq data (HIO)	Holloway et al. (2020)	ArrayExpress: E-MTAB-9228
Raw scRNA-seq data (HIO)	Yu et al. (2021)	ArrayExpress: E-MTAB-10187
Raw scRNA-seq data (HIO)	Yu et al. (2021)	ArrayExpress: E-MTAB-10268
Raw scRNA-seq data (HIO)	This study	ArrayExpress: E-MTAB-11338
Raw scRNA-seq data (HIO)	This study	ArrayExpress: E-MTAB-11347
Experimental models: Cell lines		
H9 ESC	WiCell	NIH registry #0062; RRID: CVCL_9773
UM 63–1 ESC	MStem Cell Laboratories	NIH registry #0277; RRID: CVCL_R782
iPSC 72.3	Cincinnati Children's Hospital	N/A
iPSC WTC11	Coriell Institute	RRID: CVCL_Y803
Oligonucleotides		
Primer: GAPDH Forward: CTCTGCTCCTCTGTTTCGAC	IDT	N/A
Primer: GAPDH Reverse: TTAAAAGCAGCCCTGGTGAC	IDT	N/A
Primer: ECAD Forward: TTGACGCCGAGAGCTACAC	IDT	N/A
Primer: ECAD Reverse: GACCGGTGCAATCTTCAAA	IDT	N/A
Primer: CDX2 Forward: GGGCTCTCTGAGAGGCAGGT	IDT	N/A
Primer: CDX2 Reverse: GGTGACGGTGGGGTTAGCA	IDT	N/A
Primer: PDX1 Forward: CGTCCGCTTGTCTCCTC	IDT	N/A
Primer: PDX1 Reverse: CCTTCCCATGGATGAAGTC	IDT	N/A
Primer: MUC2 Forward: TGTAGGCATCGCTCTTCTCA	IDT	N/A
Primer: MUC2 Reverse: GACACCATCTACCTCACCCG	IDT	N/A
See Table S2 for all other primer sequences	IDT	N/A
Software and algorithms		
ImageJ	Schneider et al. (2012)	https://imagej.nih.gov/ij/
Prism 8.3.0	GraphPad	https://www.graphpad.com/scientific-software/prism/
Scanpy, Ingest	(Wolf et al., 2018)	https://github.com/theislab/scanpy
BBKNN	(Polański et al., 2020)	https://github.com/Teichlab/bbknn
UMAP	(McInnes et al., 2018)	https://github.com/lmcinnes/umap
Python 3.7.3	Python	Python.org
Detailed methods and code for scRNAseq analysis	GitHub	https://github.com/jason-spence-lab/Capeling_2022
Other		
Matrigel	Corning	Cat#354234
Alginate	Alfa Aesar	Cat# B25266
Histoclear II	National Diagnostics	Cat#HS-202